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AN EXTREME VALUES OF THE FUNCTION $S(T)$ IN SHORT INTERVALS[1]

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We obtain new estimates for the maximum and minimum of the argument of the Riemann zeta-function on very short segments of the critical line. These results are based on the Riemann hypothesis.

Key words : Riemann zeta-function; extreme values; critical line.

1. INTRODUCTION

First we give some necessary definitions.

For a real t distinct from the ordinate of any zero of the Riemann zeta function $\zeta(s)$, we set

$$S(t) \,=\, \pi^{-1}\,\arg\zeta\big(\tfrac{1}{2}+it\big),$$

where $\arg\zeta\big(\frac{1}{2}+it\big)$ is obtained by the continuous extension of continuous branch of $\arg\zeta(s)$ along the broken line starting at the point $s=0$ ($\arg\zeta(2)=0$) and then going first to the point $s=2+it$ and then to $s=\frac{1}{2}+it$. If t is the imaginary part of a zero of $\zeta(s)$ then we set $S(t)=S(t+0)$.

Next, for a positive t which differs from the imaginary part of any zero of $\zeta(s)$, we denote by $N(t)$ the number of zeros of the zeta function in the rectangle $0\le\operatorname{Re}s\le 1$, $0<\operatorname{Im}s\le t$. If t coincides with the imaginary part of a zero then we set $N(t)=N(t+0)$.

The following Riemann - von Mangoldt formula

$$N(t) \,=\, \frac{t}{2\pi}\log\frac{t}{2\pi} \,-\, \frac{t}{2\pi} \,+\, \frac{7}{8} \,+\, S(t) \,+\, O(t^{-1}), \quad t\to+\infty,$$

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shows that the function $S(t)$ is, in essence, the remainder term in the expression for $N(t)$ (see, for example, [1, Ch. IX, §3]). This is the reason why the behavior of this function is very interesting for the number-theorists.

In particular, the problems concerning upper bounds for $|S(t)|$ and extreme values of $S(t)$ are of special interest. In 1904, Mangoldt [2] proved that $S(t) = O(\log t)$ as $t \to +\infty$, and this result is still the best (in the sense of order of growth; for the further history of this and related questions, see the survey [3]). The Riemann hypothesis allows one to replace $\log t$ by $(\log t)(\log\log t)^{-1}$ (for the best value of the constant in O-symbol, see [4]). In [5], one can find a series of conjectures related the behavior of maximum of $\zeta(\frac{1}{2} + it)$.

At the same time, it is known that

$$\sup_{T\le t\le 2T} \big\{\pm S(t)\big\} \;\ge\; c\bigg(\frac{\log T}{\log\log T}\bigg)^{a} \tag{1}$$

where $a = \frac{1}{3}$ unconditionally [6] and $a = \frac{1}{2}$ under the Riemann hypothesis [7] (see also the pioneering paper of Selberg [8]).

To improve (1), one can try to replace the segment $T \le t \le 2T$ by shorter one: $T \le t \le T + H$, $H = o(T)$. Thus, in [3], the same result for the case $H = T^{\alpha+\varepsilon}$ where $\alpha = \frac{27}{82}$, is established. If we assume the Riemann hypothesis, we can prove the analogs of (1) for extremely short segments. Thus, in [9] we find that

$$\sup_{T-H\le t\le T+2H} \big\{\pm S(t)\big\} \;\ge\; \frac{1}{90\pi}\sqrt{\frac{\log H}{\log\log H}}$$

for any $T > T_0$ and $(\log T)(\log\log T)^{-3/2} \le H \le T$. Later, Boyarinov [10] improved this result (in the sense of length H) and established the bound

$$\sup_{T-H\le t\le T+2H} \big\{\pm S(t)\big\} \;\ge\; \frac{1}{900}\frac{\sqrt{\log H}}{\log\log H}$$

for $\sqrt{\log\log T} \le H \le (\log T)(\log\log T)^{-3/2}$.

The aim of the present paper is to prove the analog of last two theorems for the case when H is essentially smaller than $\sqrt{\log\log T}$. Namely, we prove

Theorem — *Let $m \ge 2$ be any fixed integer, $T > T_0(m) > m$ and let*

$$\frac{2(2m\log\log T)^{\frac{1}{2m}}}{\log\log\log\log T} \;\le\; H \;\le\; \sqrt{\log\log T}.$$

If the Riemann hypothesis is true then the inequalities

$$\sup_{T-H\le t\le T+2H}\{\pm S(t)\} \ge \frac{1}{50\pi}\,\frac{\sqrt{\log H}}{(8m\log\log H)^m}$$

hold.

Notations : We use the following notations:

– $m \ge 2$ is any fixed integer;

– $\Phi(u) = \exp\Big(-\frac{u^{2m}}{2m}\Big)$;

– $\Lambda(n)$ denotes von Mangoldt's function: $\Lambda(n) = \log p$ if $n = p^m$ and p is prime, and $\Lambda(n) = 0$ otherwise;

– $\widehat{f}$ denotes the Fourier transform of f, that is $\widehat{f}(\lambda) = \int\limits_{-\infty}^{+\infty} f(u)e^{-i\lambda u}du$;

– $\theta, \theta_1, \theta_2, \dots$ denote complex numbers whose absolute values do not exceed one, different in different relations.

We need some auxilliary assertions.

Lemma 1 — The function $\widehat{\Phi}(\lambda)$ decreases monotonically on the segment $0 \le \lambda \le 1$ from the value $\widehat{\Phi}(0) = \frac{2\Gamma(1/(2m))}{(2m)^{1-1/(2m)}}$ to the value $\widehat{\Phi}(1) > \frac{5}{4}\,\Phi\Big(\frac{\pi}{4}\Big)$. Moreover, there exists the constant $\lambda_0 = \lambda_0(m)$ such that the inequality

$$\big|\widehat{\Phi}(\lambda)\big| < \frac{5}{\sqrt{m}}\,|\lambda|^{-\beta}\exp\Big(-\frac{|\lambda|^\alpha}{\alpha}\sin(\pi\kappa)\Big)$$

holds for any real λ, $|\lambda| > \lambda_0$, with

$$\alpha = \frac{2m}{2m-1}, \quad \beta = \frac{m-1}{2m-1}, \quad \kappa = \frac{1}{2(2m-1)}.$$

PROOF : Differentiating the equation

$$\widehat{\Phi}(\lambda) = 2\int\limits_{-\infty}^{+\infty}\Phi(u)\cos(\lambda u)du$$

with respect to λ, we obtain $\widehat{\Phi}'(\lambda) = -2j(\lambda)$ where

$$j(\lambda) = \int\limits_{-\infty}^{+\infty} u\Phi(u)\sin(\lambda u)du.$$

Suppose that $\lambda > 0$. Using the inequalities $\sin(\lambda u) \geq \frac{2}{\pi}\,\lambda u$ for $0 \leq u \leq \frac{\pi}{2\lambda}$ and $\sin(\lambda u) \geq 0$ for $\frac{\pi}{2\lambda} \leq u \leq \frac{\pi}{\lambda}$ and splitting the integral $j(\lambda)$ into the sum

$$\Big(\int\limits_0^{\frac{\pi}{2\lambda}} + \int\limits_{\frac{\pi}{2\lambda}}^{\frac{\pi}{\lambda}} + \int\limits_{\frac{\pi}{\lambda}}^{+\infty}\Big)\, u\Phi(u)\sin(\lambda u)\,du \;=\; j_1 + j_2 + j_3,$$

we get

$$j_1 \;\geq\; \frac{2\lambda}{\pi}\int\limits_0^{\frac{\pi}{2\lambda}} u^2\Phi(u)du \;>\; \frac{2\lambda}{\pi}\,\Phi\Big(\frac{\pi}{2\lambda}\Big)\int\limits_0^{\frac{\pi}{2\lambda}} u^2du \;=\; \frac{1}{3}\Big(\frac{\pi}{2\lambda}\Big)^2\Phi\Big(\frac{\pi}{2\lambda}\Big), \quad j_2 \;>\; 0;$$

$$|j_3| \;\leq\; \int\limits_{\frac{\pi}{\lambda}}^{+\infty} u\Phi(u)du \;=\; (2m)^{\frac{1}{m}-1}\int\limits_{\frac{1}{2m}\left(\frac{\pi}{\lambda}\right)^{2m}}^{+\infty} w^{\frac{1}{m}-1}e^{-w}dw$$

$$<\; \Big(\frac{\lambda}{\pi}\Big)^{2(m-1)}\int\limits_{\frac{1}{2m}\left(\frac{\pi}{\lambda}\right)^{2m}}^{+\infty} e^{-w}dw \;=\; \Big(\frac{\lambda}{\pi}\Big)^{2(m-1)}\Phi\left(\frac{\pi}{\lambda}\right).$$

Hence,

$$j(\lambda) \;>\; j_1 - |j_3| \;>\; \frac{1}{3}\Big(\frac{\pi}{2\lambda}\Big)^2\Phi\Big(\frac{\pi}{2\lambda}\Big)\Big\{1 \;-\; 12\Big(\frac{\lambda}{\pi}\Big)^{2m}\exp\Big(-\frac{1}{2m}\,(1-2^{-2m})\Big(\frac{\pi}{\lambda}\Big)^{2m}\Big)\Big\}.$$

If $0 < \lambda \leq 1$ then the expression in the figure brackets is bounded from below by the value

$$1 \;-\; \frac{12}{\pi^4}\,\exp\Big(-\frac{15}{4}\Big(\frac{\pi}{2}\Big)^4\Big) \;>\; 1 - 2\cdot 10^{-10} \;>\; 0$$

uniformly for $m \geq 2$. Therefore, $\Phi'(\lambda) < 0$ for $0 < \lambda \leq 1$. Thus we prove the first assertion of the lemma.

Next, splitting the expression for $\widehat{\Phi}(1)$ into the sum

$$2\int\limits_{-\infty}^{+\infty}\Phi(u)\cos u\,du \;=\; 2\Big(\int\limits_0^{\frac{\pi}{4}} + \int\limits_{\frac{\pi}{4}}^{\frac{\pi}{2}} + \int\limits_{\frac{\pi}{2}}^{+\infty}\Big)\Phi(u)\cos u\,du \;=\; 2(j_1 + j_2 + j_3)$$

and using the same arguments as above, we obtain:

$$j_1 \,>\, \Phi\Big(\frac{\pi}{4}\Big) \int\limits_0^{\frac{\pi}{4}} \cos u\, du \,=\, \frac{1}{\sqrt{2}}\Phi\Big(\frac{\pi}{4}\Big), \quad j_2 \,>\, 0,$$

$$|j_3| \,\le\, (2m)^{\frac{1}{2m}-1} \int\limits_{\frac{1}{2m}\left(\frac{\pi}{2}\right)^{2m}}^{+\infty} w^{\frac{1}{2m}-1} e^{-w} dw \,\le\, \left(\frac{2}{\pi}\right)^{2m} \Phi\Big(\frac{\pi}{2}\Big),$$

and hence

$$\widehat{\Phi}(1) \,>\, j_1 - |j_3| \,>\, 2\Big\{ \frac{1}{\sqrt{2}}\, \Phi\Big(\frac{\pi}{4}\Big) \,-\, \left(\frac{2}{\pi}\right)^4 \Phi\Big(\frac{\pi}{2}\Big) \Big\}$$
$$= \,\sqrt{2}\, \Phi\Big(\frac{\pi}{4}\Big) \Big\{ 1 \,-\, \sqrt{2} \Big(\frac{2}{\pi}\Big)^4 \frac{\Phi\Big(\frac{\pi}{2}\Big)}{\Phi\Big(\frac{\pi}{4}\Big)} \Big\}.$$

One can note that

$$\frac{\Phi\Big(\frac{\pi}{2}\Big)}{\Phi\Big(\frac{\pi}{4}\Big)} \,=\, \exp\Big\{ -\frac{1}{2m}\,(1 - 2^{-2m}) \Big(\frac{\pi}{2}\Big)^{2m} \Big\} \,\le\, \exp\left(-\frac{15}{64} \Big(\frac{\pi}{2}\Big)^4 \right)$$

for any $m \ge 2$. Thus we arrive at the desired bound for $\widehat{\Phi}(1)$.

Finally, the last assertion of the lemma follows from the formula

$$\int\limits_{-\infty}^{+\infty} \exp\left(-\frac{u^{2m}}{2m} + i\lambda u \right) du$$
$$= \,\frac{4\sqrt{\pi\kappa}}{\lambda^\beta}\, \exp\left(-\frac{\lambda^\alpha}{\alpha}\, \sin(\pi\kappa) \right) \Big\{ \cos\left(\frac{\lambda^\alpha}{\alpha}\, \cos(\pi\kappa) \right) \,+\, O\big(\lambda^{-\alpha}\big) \Big\},$$

where $\lambda \to +\infty$ and α, β, κ are defined as above (see [11, §7.1]).

Lemma 2 — The following inequalities hold true:

$$|S(t)| \,\le\, \begin{cases} 1, & \textit{if } \ |t| \le 280, \\ 1.05 \log|t|, & \textit{if } \ |t| > 280. \end{cases}$$

PROOF : The first inequality follows from the data of Table 1 from [12] and the second one follows from the classical estimate of Backlund [13]:

$$|S(t)| \,<\, 0.1361\log|t| \,+\, 0.4422\log\log|t| \,+\, 4.3451 \,\leq$$
$$\leq\, (\log|t|)\left(0.1361 \,+\, 0.4422\,\frac{\log\log 280}{\log 280} \,+\, \frac{4.3451}{\log 280}\right) \,<\, 1.05\log|t|.$$

The lemma is proved.

Let $\tau > 1$ and $f(u) = \Phi(\tau u)$.

Lemma 3 — If the Riemann hypothesis is true then the formula

$$\int\limits_{-\infty}^{+\infty} f(u)S(t+u)du \,=\, -\,\frac{1}{\pi}\sum_{n=2}^{+\infty}\frac{\Lambda(n)}{\sqrt{n}}\,\frac{\sin(t\log n)}{\log n}\,\widehat{f}(\log n) \,-\, 2\int\limits_{0}^{1/2} f(-t-iu)du$$

holds.

PROOF : The proof of this assertion repeats word -by -word (with the minor changes) the proof of Theorem 3 from [3]. The difference is that we need to use the explicit formula for $f(z)$ instead of the inequality $|f(z)| \leq c(|z|+1)^{-(1+\alpha)}$.

Lemma 4 — Let $y > y_0 > 0$, $\mu, \nu \geq 0$, $k \geq 1$, $k = \mu+\nu$, and let $p_1, \ldots, p_\mu$, $q_1, \ldots, q_\nu$ range over the primes in the interval $(1, y]$ that satisfy the condition $p_1 \ldots p_\mu \neq q_1 \ldots q_\nu$. If $a(p)$ is a sequence of complex numbers in which $|a(p)| \leq \delta$ for any prime $p \leq y$, then the integral

$$I \,=\, \int\limits_{T}^{T+H} \sum_{\substack{p_1,\ldots,p_\mu\\ q_1,\ldots,q_\nu}} \frac{a(p_1)\ldots a(p_\mu)\,\overline{a}(q_1)\ldots\overline{a}(q_\nu)}{\sqrt{p_1\ldots q_\mu}}\left(\frac{p_1\ldots p_\mu}{q_1\ldots q_\nu}\right)^{it} dt$$

satisfies the estimate $|I| \leq (\delta^2 y^3)^k$.

PROOF : It is lemma 2 from [14, §2.1].

Lemma 5 — Let $k \geq 1$ be an integer, let $M > 0$ and let a real function $W(t)$ satisfies the inequalities

$$\int\limits_{T}^{T+H} W^{2k}(t)\,dt \,\geq\, HM^{2k}, \quad \left|\int\limits_{T}^{T+H} W^{2k+1}(t)\,dt\right| \,\leq\, \frac{1}{2}\,HM^{2k+1}.$$

Then

$$\sup_{T\leq t\leq T+H}\big\{\pm W(t)\big\} \,\geq\, \frac{M}{2}.$$

PROOF : It is a slight modification of lemma 4 from the paper of Tsang [6].

PROOF OF THE THEOREM : Let $\tau = 2\log\log H$ and suppose that $T \le t \le T + H$. Applying lemma 3 to the function $f(u) = \Phi(\tau u)$ we obtain

$$\tau \int\limits_{-\infty}^{+\infty} \Phi(\tau u) S(t+u)du \;=\; -\frac{1}{\pi}\, V(t) \;+\; R_1(t),$$

where

$$V(t) = \sum_{n=2}^{+\infty} \frac{\Lambda(n)}{\sqrt{n}}\, \frac{\sin\left(t\log n\right)}{\log n}\, \widehat{\Phi}\Big(\frac{\log n}{\tau}\Big), \quad R_1(t) = 2\int\limits_0^{1/2} \Phi\big(\tau(t+iu)\big)\, du.$$

Since

$$\mathrm{Re}\,(t+iu)^{2m} \;=\; t^{2m}\left(1 \;+\; \sum_{\nu=1}^{m} (-1)^{\nu} \binom{2m}{2\nu} \left(\frac{u}{t}\right)^{2m}\right)$$

and since the absolute value of the last sum is bounded from above by the values

$$\sum_{\nu=1}^{2m} \binom{2m}{\nu} (2t)^{-\,2\nu} \;=\; \left(1 \;+\; \frac{1}{2t}\right)^{2m} - 1 \;\le\; \frac{1}{2t}\left(1 + \frac{1}{2t}\right)^{2m-1} \;<\; \frac{e}{2t},$$

we find that

$$\big|\Phi\big(\tau(t+iu)\big)\big| \;=\; \exp\left\{-\,\frac{\tau^{2m}}{2m}\;\mathrm{Re}(t+iu)^{2m}\right\} \;<\; \exp\left\{-\,\frac{(\tau t)^{2m}}{8m}\right\},$$

$$|R_1(t)| \;\le\; 2\cdot\frac{1}{2}\,\exp\left\{-\,\frac{(\tau t)^{2m}}{8m}\right\}.$$

Further, let $X = \exp\big((4m\tau)^{2m}\big)$ and let $R_2(t)$ be the contribution to the sum $V(t)$ from the terms with $n > X$. Since $\frac{\log n}{\tau} > \lambda_0$ for any $n > X$ (λ_0 is defined in lemma 1) then lemma 1 implies that

$$\left|\widehat{\Phi}\Big(\frac{\log n}{\tau}\Big)\right| \;\le\; \frac{5}{\sqrt{m}}\left(\frac{\tau}{\log n}\right)^{\beta} \exp\left\{-\,\frac{\sin\left(\pi\kappa\right)}{\alpha}\left(\frac{\log n}{\tau}\right)^{\alpha}\right\}, \tag{1}$$

where the values α, β and κ are defined above. By the inequality $\frac{\sin\left(\pi\kappa\right)}{\alpha} \;\ge\; \frac{2}{\pi}\,\frac{\pi\kappa}{\alpha} \;=\; \frac{1}{2m}$ we conclude from (1) that

$$|R_2(t)| \;<\; \frac{5}{\sqrt{m}}\left(\frac{\tau}{\log X}\right)^{\beta} \sum_{n>X} \frac{1}{\sqrt{n}}\, \exp\left\{-\,\frac{1}{2m}\left(\frac{\log n}{\tau}\right)^{\alpha}\right\}.$$

In order to estimate the last sum,we take $r_0 = \Big[\frac{\log X}{\tau}\Big] = \big[(4m)^{2m}\tau^{2m-1}\big]$ and split the domain of summation into the segments of the type $e^{r\tau} < n \le e^{(r+1)\tau}$, $r = r_0, r_0+1, \ldots$. The sum over

such segment is bounded by the values

$$\exp\left\{-\frac{r^\alpha}{2m}\right\}\sum_{e^{r\tau}<n\le e^{(r+1)\tau}}\frac{1}{\sqrt{n}} < 2\exp\left\{\frac{(r+1)\tau}{2}-\frac{r^\alpha}{2m}\right\}$$
$$< \exp\left\{r\tau-\frac{r^\alpha}{2m}\right\} < \exp\left\{-\frac{r^\alpha}{4m}\right\}.$$

Hence,

$$|R_2(t)| < \frac{5}{\sqrt{m}}\Big(\frac{\tau}{\log X}\Big)^\beta\int\limits_{(4m\tau)^{2m-1}}^{+\infty}\exp\left\{-\frac{u^\alpha}{2m}\right\}du$$
$$= \frac{5}{\sqrt{m}}\Big(\frac{\tau}{(4m\tau)^{2m}}\Big)^\beta\frac{(2m)^{\frac{1}{\alpha}}}{\alpha}\int\limits_{\frac{1}{2m}(4m\tau)^{2m}}^{+\infty}w^{\frac{1}{\alpha}-1}e^{-w}dw$$
$$< \frac{5}{\sqrt{m}}\Big(\frac{\tau}{(4m\tau)^{2m}}\Big)^\beta\frac{1}{2\tau}\exp\left\{-\frac{(4m\tau)^{2m}}{2m}\right\}$$
$$= \frac{5(4m)^{\frac{1}{2(2m-1)}}}{(4m\tau)^m}\exp\left\{-\frac{(4m\tau)^{2m}}{2m}\right\} \le \frac{5\sqrt{2}}{(4m\tau)^m}\exp\left\{-\frac{(4m\tau)^{2m}}{2m}\right\}.$$

Finally, let $R_3(t)$ be the sum over $n=p^\nu\le X$ with $\nu\ge 2$. Then the obvious estimate

$$\left|\widehat{\Phi}\Big(\frac{\log n}{\tau}\Big)\right| = \left|\int\limits_{-\infty}^{+\infty}\Phi(u)n^{-\frac{iu}{\tau}}\,du\right| \le \widehat{\Phi}(0)$$

yields

$$|R_3(t)| \le \widehat{\Phi}(0)\sum_{\nu=2}^{+\infty}\sum_{p\le X^{1/\nu}}\frac{p^{-\nu/2}}{\nu} \le \widehat{\Phi}(0)\Big(\frac{1}{2}\sum_{p\le\sqrt{X}}\frac{1}{p}+\frac{1}{3}\sum_{\nu=3}^{+\infty}\frac{p^{-\nu/2}}{\nu}\Big)$$
$$< \widehat{\Phi}(0)\Big(\frac{1}{2}\log\log X + c\Big) = \widehat{\Phi}(0)\big(m\log(4m\tau)+c\big),$$

where $c>0$ is a sufficiently large absolute constant.

Thus we get

$$\tau\int\limits_{-\infty}^{+\infty}\Phi(\tau u)S(t+u)\,du = -\frac{1}{\pi}W(t)+\theta_1Q_1,$$

where

$$W(t) = \sum_{p\le x}\frac{a(p)}{\sqrt{p}}\sin(t\log p),\quad a(p) = \Phi\Big(\frac{\log p}{\tau}\Big),$$

$$Q_1 = \exp\Big\{-\frac{(\tau t)^{2m}}{8m}\Big\} + \frac{5\sqrt{2}}{(4m\tau)^m}\exp\Big\{-\frac{(4m\tau)^{2m}}{2m}\Big\} + \widehat{\Phi}(0)\big(m\log(4m\tau)+c\big)$$
$$< \widehat{\Phi}(0)\big(m\log(4m\tau)+2c\big).$$

Now let us consider the integrals

$$I_1 = \int\limits_H^{+\infty}\Phi(\tau u)S(t+u)\,du,\quad I_2 = \int\limits_{-\infty}^{-H}\Phi(\tau u)S(t+u)\,du.$$

Splitting I_1 into the sum

$$I_1 = \Big(\int\limits_H^T + \int\limits_T^{+\infty}\Big)\Phi(\tau u)S(t+u)\,du = I_1^{(1)} + I_1^{(2)}$$

and applying lemma 2 we obtain:

$$|I_1^{(1)}| \le 1.05\int\limits_H^T\Phi(\tau u)\log(t+u)\,du < \frac{1.1\log T}{\tau}\int\limits_{H\tau}^{+\infty}\Phi(v)\,dv \le \frac{1.1}{\tau}\,\frac{\Phi(H\tau)\log T}{(H\tau)^{2m-1}},$$

$$|I_1^{(2)}| \le 1.05\int\limits_T^{+\infty}\Phi(\tau u)\log(t+u)\,du < \frac{1.1}{\tau}\int\limits_{T\tau}^{+\infty}\Phi(v)\log v\,dv \le \frac{1.1}{\tau}\,\frac{2m\Phi(T\tau)\log(T\tau)}{(T\tau)^{2m-1}},$$

and hence

$$|I_1| < \frac{1.2}{\tau}\,\frac{\Phi(H\tau)\log T}{(H\tau)^{2m-1}}.$$

Next, we split the integral I_2 into the sum

$$\int\limits_H^{+\infty}\Phi(\tau u)S(t-u)\,du = \Big(\int\limits_H^{t-10^2} + \int\limits_{t-10^2}^{t+10^2} + \int\limits_{t+10^2}^{+\infty}\Big)\Phi(\tau u)S(t-u)\,du$$
$$= I_2^{(1)} + I_2^{(2)} + I_2^{(3)}.$$

Applying the first inequality of lemma 2 to $I_2^{(2)}$ and the second one to the estimation of $I_2^{(1)}$ and $I_2^{(3)}$ we obtain

$$|I_2| < \frac{1.2}{\tau}\,\frac{\Phi(H\tau)\log T}{(H\tau)^{2m-1}},\quad |I_1|+|I_2| < \frac{2.4}{\tau}\,\frac{\Phi(H\tau)\log T}{(H\tau)^{2m-1}}.$$

Finally, we have

$$j(t) \,=\, \tau \int\limits_{-H}^{H} \Phi(\tau u) S(t+u)\, du \,=\, -\,\frac{1}{\pi}\, W(t) \,+\, \theta\, Q_2,$$

where

$$Q_2 \,=\, \widehat{\Phi}(0)\big(m \log{(4m\tau)} \,+\, 2c\big) \,+\, \frac{2.4}{\tau}\, \frac{\Phi(H\tau) \log T}{(H\tau)^{2m-1}}.$$

Since $H\tau > (2m \log\log T)^{\frac{1}{2m}}$ for H and m under considering, we find that

$$\Phi(H\tau) \,<\, (\log T)^{-1}, \quad Q_2 \,<\, 2\widehat{\Phi}(0) m \log{(4m\tau)}.$$

Now let us take $k = \Big[\dfrac{\log H}{5 \log X}\Big]$ and define the integrals $I(k)$ and $J(k)$ by the following relations:

$$I(k) \,=\, \int\limits_{T}^{T+H} W^{2k}(t)\, dt, \quad J(k) \,=\, \int\limits_{T}^{T+H} W^{2k+1}(t)\, dt.$$

Writing $W(t)$ as

$$\frac{1}{2i}\, \big(U(t) \,-\, \overline{U}(t)\big), \quad U(t) \,=\, \sum_{p \le X} \frac{a(p)}{\sqrt{p}}\, p^{it},$$

we find that

$$I(k) \,=\, (2i)^{-2k} \sum_{\nu=0}^{2k} (-1)^{\nu} \binom{2k}{\nu} j_{\nu}, \quad j_{\nu} \,=\, \int\limits_{T}^{T+H} U^{\,\nu}(t) \overline{U}^{\,2k-\nu}(t) dt.$$

The application of lemma 4 with $\delta = \widehat{\Phi}(0)$ to the case $\nu \ne k$ yields:

$$|j_{\nu}| \,<\, \Big(\widehat{\Phi}(0) X^{\frac{3}{2}}\Big)^{2k} < X^{4k} \le H^{\frac{4}{5}}.$$

The same estimate is valid for the contribution to j_k of the terms under the condition $p_1 \dots p_k \ne q_1 \dots q_k$. Hence,

$$I(k) \,=\, 2^{-2k} \binom{2k}{k} H \mathfrak{S}_k \,+\, \theta H^{\frac{4}{5}},$$

where

$$\mathfrak{S}_k \,=\, \sum_{\substack{p_1 \dots p_k = q_1 \dots q_k \\ p_1, \dots, q_k \le X}} \frac{a^2(p_1) \dots a^2(p_k)}{p_1 \dots p_k}.$$

In order to estimate the sum $\mathfrak{S}_k$ from the below, we truncate the sum by replacing the upper bound for $p_1, \ldots, q_k$ by the value $Y = e^{\tau} = (\log H)^2 < X$. Then it follows from lemma 1 that the inequalities

$$a(p) \,=\, \widehat{\Phi}\Big(\frac{\log p}{\tau}\Big) \,\geq\, \widehat{\Phi}(1)$$

hold for any $p \leq Y$. Further, if we retain the terms in the truncated sum that correspond to the tuples $(p_1, \ldots, p_k)$ involving no repetitions and using the fact that the number of solutions $(q_1, \ldots, q_k)$ of the equation $p_1 \ldots p_k = q_1 \ldots q_k$ is equal to $k!$, we obtain that

$$\mathfrak{S}_k \,=\, \big(\widehat{\Phi}(1)\big)^{2k} \sum_{\substack{p_1 \ldots p_k = q_1 \ldots q_k \\ p_1, \ldots, q_k \leq Y}} (p_1 \ldots p_k)^{-1} \,\geq\, k!\,\big(\widehat{\Phi}(1)\big)^{2k} \sum_{\substack{p_1, \ldots, p_k \leq Y \\ p_1, \ldots, p_k \text{ are distinct}}} (p_1 \ldots p_k)^{-1}.$$

Applying the arguments from [9] (the estimate of the sum Σ) to the estimation of the last sum, we find that

$$\mathfrak{S}_k \,\geq\, k!\,\big(\widehat{\Phi}(1)\big)^{2k} \bigg(\sum_{2k \log k < p \leq Y} \frac{1}{p} \bigg)^k \,\geq\, k!\,\big(\widehat{\Phi}(1)\big)^{2k} \bigg(\sum_{\sqrt{Y} < p \leq Y} \frac{1}{p} \bigg)^k \,>\, k! \left(\frac{4}{5}\,\widehat{\Phi}(1) \right)^{2k},$$

$$I(k) \,>\, \frac{(2k)!}{k!}\,\frac{H}{2^{2k}} \left(\frac{4}{5}\,\widehat{\Phi}(1) \right)^{2k} \,-\, H^{\frac{4}{5}} \,>\, \frac{e}{2} \bigg(\frac{4k}{e} \bigg)^k \bigg(\frac{\widehat{\Phi}(1)}{5} \bigg)^{2k} \,-\, H^{\frac{4}{5}} \,>\, HM^{2k},$$

where

$$M \,=\, \frac{2}{5}\,\widehat{\Phi}(1) \sqrt{\frac{k}{e}} \,>\, 2.$$

Finally, Lemma 4 yields:

$$|J(k)| \,<\, \Big(\widehat{\Phi}(0) X^{\frac{3}{2}} \Big)^{2k+1} \,<\, X^{4k} \leq H^{\frac{4}{5}} \,<\, \frac{1}{2}\,HM^{2k+1}.$$

Now it follows from lemma 5 that there exist the values t_0 and t_1 such that $T \leq t_0, t_1 \leq T + H$ such that $W(t_0) < -0.5M$ and $W(t_1) > 0.5M$. Thus we have

$$j(t_0) \,>\, -\frac{1}{\pi}\,W(t_0) \,-\, Q_2 \,>\, \frac{M}{2\pi} \,-\, 2\widehat{\Phi}(0) m \log(4m\tau),$$
$$j(t_1) \,<\, -\frac{1}{\pi}\,W(t_1) \,+\, Q_2 \,<\, -\frac{M}{2\pi} \,+\, 2\widehat{\Phi}(0) m \log(4m\tau).$$

Setting $M_j = \sup\limits_{|u| \leq H} (-1)^j S(t_j + u)$ for $j = 0, 1$, we obviously have

$$j(t_0) \,<\, M_0 \tau \int\limits_{-H}^{H} \Phi(\tau u)\, du \,<\, M_0 \widehat{\Phi}(0), \quad j(t_1) \,>\, M_1 \widehat{\Phi}(0)$$

and therefore

$$(-1)^j M_j \,>\, \mu, \quad \mu \,=\, \widehat{\Phi}(0)\,\frac{M}{2\pi} \,-\, 2m\log\,(4m\tau).$$

Finally, applying Lemma 1 together with the inequality

$$\Gamma\biggl(\frac{1}{2m}\biggr) \,=\, 2m\Gamma\biggl(1+\frac{1}{2m}\biggr) \,\le\, 2m,$$

we obtain

$$\begin{aligned}
\mu \,&\ge\, \frac{1}{5\pi}\,\frac{\widehat{\Phi}(1)}{\widehat{\Phi}(0)}\,\sqrt{\frac{k}{e}} \,-\, 2m\log\,(4m\tau) \\
&>\, \frac{1}{8\pi}\,\frac{\Phi\Bigl(\frac{\pi}{4}\Bigr)}{\Gamma\Bigl(\frac{1}{2m}\Bigr)}\,(2m)^{1-\frac{1}{2m}}\sqrt{\frac{1}{e}\biggl(\frac{\log H}{5(4m\tau)^{2m}} \,-\, 1\biggr)} \,-\, 2m\log\,(4m\tau) \\
&>\, \frac{1}{32\pi}\,\frac{\Phi\Bigl(\frac{\pi}{4}\Bigr)}{2m}\,(2m)^{1-\frac{1}{2m}}\,\frac{\sqrt{\log H}}{(4m\tau)^m} \\
&\ge\, \frac{1}{32\pi\sqrt{2}}\,\exp\biggl\{-\frac{1}{4}\Bigl(\frac{\pi}{4}\Bigr)^4\biggr\}\,\frac{\sqrt{\log H}}{(8m\log\log H)^m} \,>\, \frac{1}{50\pi}\,\frac{\sqrt{\log H}}{(8m\log\log H)^m}.
\end{aligned}$$

The theorem is proved.

Remark 1 : The assertion of the theorem can be generalized to the case when m grows with T.

Remark 2 : During the process of publishing, some new theorems concerning the extreme values of the absolute value and the argument of $\zeta(s)$ on the extremely short segments of the critical line $\operatorname{Re} s = \frac{1}{2}$ were proved (see [15-17]).

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REFERENCES

1. E. C. Titchmarsh, The theory of the Riemann Zeta -function, 2nd ed. revised by D. R. Heath -Brown, Clarendon Press, Oxford, 1986.
2. H. V. Mangoldt, Die Nullstellen der Riemannschen funktion $\xi(t)$, *Math. Ann.*, **60**(1) (1905), 1-19.
3. A. A. Karatsuba and M. A. Korolev, The argument of the Riemann zeta function, *Russian Math. Surveys*, **60**(3) (2005), 433-488.

4. E. Carneiro, V. Chandee and M. B. Milinovich, Bounding $S(t)$ and $S_1(t)$ on the Riemann hypothesis, *Math. Ann.*, **356**(3) (2013), 939-968.

5. D. W. Farmer, S. M. Gonek and C. P. Hughesm, The maximum size of L-functions, *J. Reine Angew. Math.*, **609** (2007), 215-236.

6. K.-M. Tsang, Some Ω-theorems for the Riemann zeta function, *Acta Arith.*, **46**(4) (1986), 369-395.

7. H. L. Montgomery, Extreme values of the Riemann zeta function, *Comment. Math. Helv.*, **52**(4) (1977), 511-518.

8. A. Selberg, Contributions to the theory of the Riemann zeta-function, *Arch. Math. Naturvid.*, **48**(5) (1946), 89-155.

9. M. A. Korolev, On large values of the function $S(t)$ on short intervals, *Izvestiya: Mathematics*, **69**(1) (2005), 113-122.

10. R. N. Boyarinov, On large values of the function $S(t)$ on short intervals, *Math. Notes*, **89**(4) (2011), 472-479.

11. M. V. Fedoryuk, *The Saddle-point method*, Nauka, Moscow, 1977 (in Russian).

12. R. Sherman Lehman, On the distribution of zeros of the Riemann zeta-function, *Proc. London Math. Soc.*, **20**(3) (1970), 303-320.

13. R. J. Backlund, Über die Nullstellen der Riemannschen zetafunktion, *Acta Math.*, **41** (1916), 345-375.

14. A. A. Karatsuba and M. A. Korolev, Behaviour of the argument of the Riemann zeta function on the critical line, *Russian Math. Surveys*, **61**(3) (2006), 389-482.

15. M. A. Korolev, On large values of the Riemann zeta -function on short segments of the critical line, *Acta Arith.*, **166**(4) (2014), 349-390.

16. M. A. Korolev, Large values of the Riemann zeta -function on short intervals of the critical line, *Dokl. Math.*, **91**(1) (2015), 102-104.

17. M. A. Korolev, On the large values of the Riemann zeta-function on the critical line - II, *Moscow J. Comb. Number Theory*, **5**(3) (2015), 60-86.